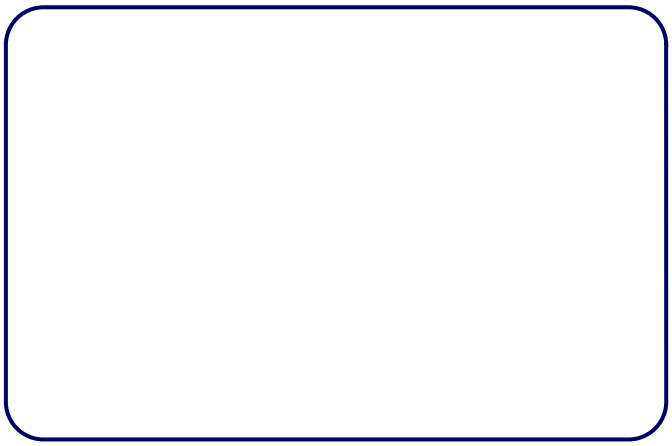
Task

1. Draw a diagram of the model in the box. Add labels around the box to show what each part represents.



2. What happens to the balloon inside when you pull down the large balloon?

.....

3. Explain your observation.

.....

.....

4. Complete the table to describe what happens to each of these structures when we inhale and exhale.

	diaphragm	ribcage	volume inside ribcage	Pressure inside ribcage	lungs
inhale	contracts				inflate
exhale				decreases	

5. Complete these sentences

i. I think my model of the lung is realistic because

.....

ii. I think the model of the lung is not realistic because

.....

iii. I think the model of the lung could be improved by

.....

Keywords

lungs	trachea	ribcage	diaphragm	contract
relax	inflate	deflate	volume	pressure

Teaching notes and answers

Students could make their own model lung or watch a demo or a video clip. For example the video 'Model of lung' (http://www.youtube.com/watch?v=CBv2BqqAydE) shows how to make a model lung and what happens as the 'diaphragm' relaxes and contracts.

1. Diagram should include the labels:

   balloon = lungs, straw = trachea, bottle = rib cage, large balloon = diaphragm

2. The balloon inflates

3. As the 'diaphragm' moves down, the volume in the bottle increases, the pressure inside decreases compared to air pressure causing the small balloon to inflate.

4.

	diaphragm	ribcage	volume inside ribcage	pressure inside ribcage	lungs
inhale	contracts	moves up and out	increases	decreases	inflate
exhale	relaxes	moves down and in	decreases	increases	deflate